

For this coding exercise, all that is needed to run the code is to run the week3_threads.py file for it to function properly. The code imports the Python module “threading” from which all thread methods needed are imported from.

There are three main functions called: functionOne, functionTwo and functionThree that each have a specific thread assigned to call them. It looks as follows:

```
def functionOne():
    print('Function One is running')

def functionTwo():
    print('Function Two is running')

def functionThree():
    print('Function Three is running')

def main():
    """
    Entry point of the program: Create, start and join threads
    """

    #Build thread objects that map to distinct functions
    one_thread = threading.Thread(
        target=functionOne,
        name="one-thread"
    )

    two_thread = threading.Thread(
        target=functionTwo,
        name="two-thread"
    )

    three_thread = threading.Thread(
        target=functionThree,
        name="three-thread"
    )
```

Each thread is named as follows: one_thread, two_thread and three_thread. Each thread is assigned with calling the function with the same number (one_thread calls functionOne,

two_thread calls functionTwo and three_thread calls functionThree). Each thread is initialized using threading.Thread() and started. It looks as follows:

```
#Build thread objects that map to distinct functions
one_thread = threading.Thread(
    target=functionOne,
    name="one-thread"
)

two_thread = threading.Thread(
    target=functionTwo,
    name="two-thread"
)

three_thread = threading.Thread(
    target=functionThree,
    name="three-thread"
)

#Start threads concurrently
one_thread.start()
two_thread.start()
three_thread.start()
```

A join() is made for the program to wait for all threads to finish execution. Output looks as follows:

```
Function One is running
Function Two is running
Function Three is running
```

The biggest difficulty probably faced while implementing the program was understanding what the expected output was and how each function was expected to contain. After understanding that each function just needed to print its name to the console, the process was straight forward.